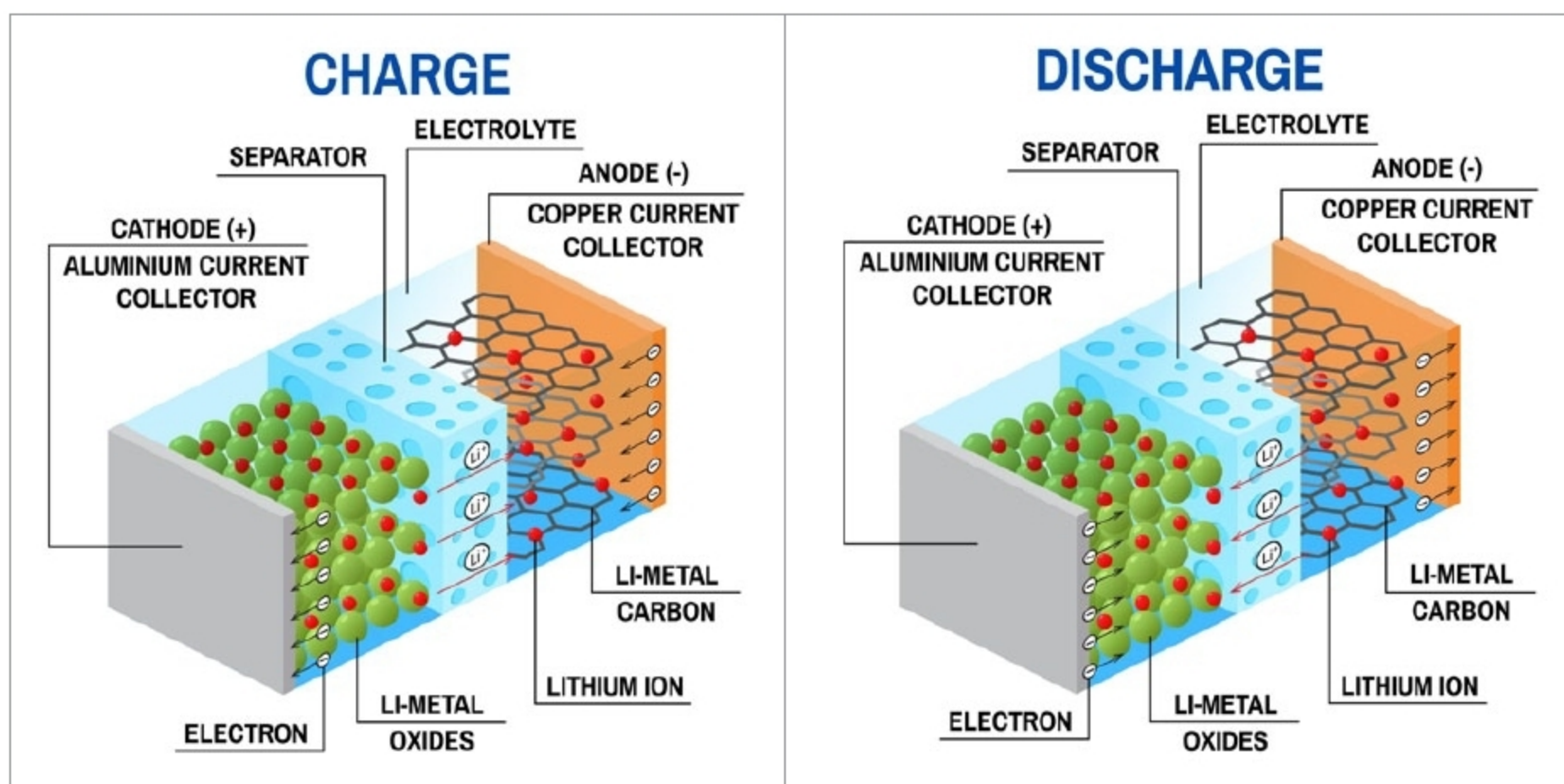


Electronics With Lithium-Ion Batteries

Lithium battery packs are particularly sensitive to faults caused by external shorts, runaway charging conditions and abusive overcharging, which can cause potentially damaging over-current and over-temperature conditions. Hence, there is a need to enforce safety regulations and established test requirements for lithium-ion and lithium-polymer packs to demonstrate their resilience to both short-circuit and overcharge events



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Nickel-cadmium had been the only suitable battery for the last many years, to be used in portable equipment from wireless communications to mobile computing. Then, other types of batteries like nickel-metal-hydride and lithium-ion emerged in the early 1990s, giving fierce competition to gain customer's acceptance. Evolution of battery technologies is directly tied to miniaturisation of mobile electronics, and to power portable electronic devices in everyday life. Lithium-ion rechargeable batteries

are extremely popular and can be found in laptops, PDAs, cellphones and iPods.

Today, lithium-ion has emerged as the fastest growing and most promising battery chemistry. Lithium-ion battery technology is further driven by consumer demands for applications that have more computing power, and come in thinner and smaller packages. In response, design engineers and components manufacturers must continue to find creative solutions that conform to these requirements.

Groundwork towards the development